

BETA-BLOCKER THERAPY

Do β -Blockers Cause Depression?Systematic Review and Meta-Analysis of Psychiatric Adverse Events During β -Blocker Therapy

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ABSTRACT: β -Blockers are important drugs in the treatment of cardiovascular diseases. They are suspected of inducing various psychiatric adverse events (PAEs), particularly depression, affecting cardiovascular morbidity and mortality. We performed a systematic search for double-blind, randomized controlled trials investigating β -blockers to analyze the risk of PAEs or withdrawal of therapy due to PAEs. We extracted the frequencies of PAEs and rates of withdrawals and reviewed them to the number of exposed patients. For β -blockers versus placebo or other active treatment, we calculated odds ratios for individual PAEs and withdrawal rates. We retrieved overall 285 eligible studies encompassing 53 533 patients. The risk of bias was judged to be high in 79% of the studies. Despite being the most frequently reported PAE with a total of 1600 cases, depression did not occur more commonly during β -blockers than during placebo (odds ratio, 1.02 [95% CI, 0.83–1.25]). β -Blocker use was also not associated with withdrawal for depression (odds ratio, 0.97 [95% CI, 0.51–1.84]). Similar results were obtained for comparisons against active agents. Among other PAEs, only unusual dreams, insomnia, and sleep disorder were possibly related to β -blocker therapy. In conclusion, this analysis of large-scale data from double-blind, randomized controlled trials does not support an association between β -blocker therapy and depression. Similarly, no effect for β -blockers was found for other PAEs, with the possible exceptions of sleep-related disorders. Consequently, concerns about β -blockers' impact on psychological health should not affect their use in clinical practice. (*Hypertension*. 2021;77:1539–1548. DOI: 10.1161/HYPERTENSIONAHA.120.16590.) • **Data Supplement**

Key Words: cardiovascular diseases ■ depression ■ heart failure ■ morbidity ■ mortality

β -Blockers are indispensable drugs in cardiovascular medicine, particularly in the management of heart failure (HF) with reduced ejection fraction, arrhythmias, angina pectoris, and hypertension.^{1–3} Although they are no longer recommended as first-line therapy in patients with uncomplicated hypertension, they are beneficial for the treatment of hypertension in specific situations, such as symptomatic angina, for heart rate control, postmyocardial infarction, HF with reduced ejection fraction, and as an alternative to angiotensin-converting enzyme inhibitors or angiotensin-receptor blockers in younger hypertensive women planning a pregnancy or of child-bearing potential.^{2,4}

Although β -blockers have been demonstrated to exert beneficial psychotropic effects in managing some

psychiatric conditions, such as anxiety⁵ and akathisia,⁶ psychiatric adverse events (PAEs) have also been reported and are related to the blockade of cerebral β -adrenergic receptors.⁷ Most importantly, β -blockers have repeatedly been linked to the emergence of depression.⁸ Furthermore, they are suspected of contributing to other conditions, including anxiety, drowsiness, insomnia, hallucinations, and nightmares.⁸ Indeed, depression and other psychiatric symptoms are frequently registered as suspected adverse events (AE) during β -blocker treatment in postmarketing surveillance systems (Table S1 in the **Data Supplement**^{9,10}). The putative association between β -blocker therapy and PAEs is highly relevant in the management of cardiovascular diseases. For instance, patients with HF or hypertension are prone to

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Novelty and Significance

What Is New?

- This is the first systematic review and meta-analysis investigating the whole spectrum of psychiatric adverse events in 53 533 individuals exposed to β -blockers in double-blind, randomized controlled parallel-arm studies; 69% were hypertension trials.
- Depression and other psychiatric adverse events occur relatively frequent during β -blocker therapy. However, they are, barring few exceptions, not related to β -blocker use.

What Is Relevant?

- β -Blockers are mostly safe regarding psychological health, and concerns about psychiatric adverse events should not affect their use in hypertension and other indications.

Summary

Various psychiatric adverse events occurred relatively frequently during β -blocker therapy, with depression being the most common. However, compared with placebo, β -blocker therapy was not adversely associated with depression and most other psychiatric adverse events.

Nonstandard Abbreviations and Acronyms

AE	adverse event
ASCOT	Anglo-Scandinavian Cardiac Outcomes Trial
HF	heart failure
PAE	psychiatric adverse event
RCT	randomized controlled trial

developing depression, and the presence of depression may negatively influence their cardiovascular morbidity and mortality.^{11–13}

However, the relationship between the use of β -blockers and the occurrence of PAEs has not yet been systematically assessed. This systematic review and meta-analysis aim to provide a current synthesis of the available data from double-blind, parallel-arm studies about the risk of PAEs associated with β -blocker use.

METHODS

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Overview

We conducted a systematic review and meta-analysis following a prespecified protocol and following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses guidelines.¹⁴ Our study was registered on PROSPERO (<https://www.crd.york.ac.uk/prospero/>).

Search Strategy

We searched for double-blind, randomized controlled trials (RCTs) with parallel-arm design indexed in the literature databases PubMed, Embase, and WebOfScience from database inception to September 11, 2020. No language

restriction was applied. The search was performed using an algorithm that combined various β -blockers with terms referring to RCTs (Table S2). We also searched the references of records identified in our search for additional trials. Third, we sought unpublished data using 6 clinical trial databases: ClinicalTrials.gov, EU Clinical Trials Register, Clinical Study DataRequest, the YODA PROJECT, and the study registers from Allergan and GlaxoSmithKline, using the β -blocker names as search terms. Fourth, we scanned the Food and Drug Administration drug approval packages of β -blockers for AE data of published and unpublished studies. Finally, we contacted authors of eligible trials that incompletely reported AE for additional data.

Study Eligibility

A detailed list of inclusion and exclusion criteria is presented in the [Data Supplement](#) (Table S3). We included double-blind, parallel-arm RCTs that compared β -blocker monotherapy with either placebo or an active comparator in adolescent or adult nonpsychiatric patients. Treatment had to be applied systemically for at least 14 days. We accepted trials with patients on a stable background medication or—in trials focusing on conditions with paroxysmal symptoms—on a rescue medication. Furthermore, we also included trials that per-protocol added other agents (backup medication) to the fraction of patients missing their therapeutic goals. Studies were only included if they reported the occurrence of at least one PAE or at least one case of withdrawal of therapy due to PAEs during either β -blocker or control treatment, or if they explicitly reported the absence of PAEs. For the meta-analysis, both placebo-controlled trials and trials using active non- β -blocking comparators were eligible. While we were mainly interested in the PAE risks of β -blockers compared with placebo, the incentive for comparison against active control was to validate our findings further.

Data Extraction

Two reviewers (T.G. Riemer and either L.E. Villagomez Fuentes or A. Bär) independently assessed trial eligibility and extracted data from eligible studies. We extracted data concerning

trial design, treatment, sample characteristics, and AE data. Regarding AE, we noted types and numbers of PAEs, AE assessment methodology, and the number of withdrawals due to PAEs. In addition to PAEs, we also extracted frequencies of fatigue/tiredness, a known AE of β -blockers, to serve as a positive control.¹⁵ Finally, we checked whether there were industry sponsorship indicators and, for trials comparing β -blockers with non- β -blocking agents, whether β -blockers were target or comparator drugs. Extracted data were reviewed for consistency by a third reviewer (either E.A.E. Algharably, L. Zaidi Touis, or M.S. Schäfer). Any disagreements were resolved through consensus.

Outcome Definitions

Our prespecified primary outcomes were frequencies of depression and withdrawal of therapy due to depression. Our secondary outcomes were frequencies of other PAEs and withdrawal due to other PAEs. We a priori defined PAE clusters with exemplary symptoms through discussion of psychiatrists and clinical pharmacologists (T.G. Riemer, M.J. Dinges, T. Majic, L.E. Villagomez Fuentes, and R. Kreutz). These symptoms were expanded with additional symptoms reported in the studies if they were judged to be PAEs by a consensus of the reviewers above (Table S4). Individual PAEs and withdrawals had to be reported as frequencies to be included. For withdrawals due to PAEs, we considered the number of patients who discontinued the study due to a PAE occurrence. Data for the positive control fatigue/tiredness was collected in the same way.

Data Synthesis

All extracted symptoms were reviewed by 3 psychiatrists (T.G. Riemer, M.J. Dinges, and T. Majic). We found that similar symptoms were reported across different studies using different terms. In such cases, symptoms were grouped into clusters (Table S4).

Quality Assessment

Eight reviewers (T.G. Riemer or L.E. Villagomez Fuentes with either E.A.E. Algharably, E. Mangelsen, L. Zaidi Touis, M.S. Schäfer, M.-A. Fürtig, or N. Bittner) rated the quality of all included studies using the Cochrane Risk of Bias tool.¹⁶ Furthermore, we used a self-designed rating instrument to assess the quality of AE measuring and reporting (Table S5). Any disagreements were resolved through discussion.

Statistical Analysis

Statistical analyses were performed using Review Manager 5.3.¹⁷ We performed pairwise meta-analyses, calculating odds ratios with 95% CI and respective *P* values for each symptom or symptom cluster reported in either at least 5 placebo-controlled trials or at least 5 active-controlled trials employing control medication of the same or similar drug classes (Table S6). To account for the heterogeneity in the study and patient characteristics, we used random-effects models. The minimum number of 5 trials was chosen, as it has been shown that 5 or more studies are required to achieve statistical power likely surpassing that of individual contributing studies in random-effect models.¹⁸ Heterogeneity in effect sizes was assessed using the

*I*² statistic. For comparisons with at least 10 contributing published studies, publication bias was evaluated by visual inspection of funnel plots.

Sensitivity Analyses

We intended to perform 6 sensitivity analyses: Selective inclusion of studies measuring frequencies of AE in a structured manner (eg, questionnaire); exclusion of the 2 studies with the greatest weight; restriction to studies on hypertensives; exclusion of studies with a high risk of bias in at least one of the categories of the Cochrane risk of bias tool; exclusion of studies with a duration of fewer than 8 weeks; and exclusion of studies with patients on backup medication.

Subgroup Analyses

We conducted subgroup analyses to explore possible sources of heterogeneity further. For placebo-controlled trials, we investigated whether the outcomes were moderated by lipophilicity of β -blockers (low, moderate, and high),¹⁹ intrinsic sympathomimetic activity (no, yes),^{20,21} indication of β -blocker therapy (hypertension, other cardiovascular diseases, noncardiovascular diseases), and industry sponsorship (presence or absence). For active-controlled trials, we performed subgroup analyses for industry sponsorship and for study incentive (β -blocker as target versus comparator drug).

RESULTS

Overview

Excluding duplicates, our literature search yielded 33 579 records. A total of 31 315 of these records were immediately rejected due to unsuitable trial design, absence of a β -blocker monotherapy group, or target population outside of inclusion criteria. Another 1979 records were excluded after the subsequent full-text screening. The most frequent reasons for exclusion were lack of either AE or PAE reporting. In total, we deemed 285 and 151 studies to be eligible for the review and meta-analyses, respectively (Figure 1).

Studies

The 285 studies included in our review encompassed 53 533 patients exposed to 24 different β -blockers. Of these, 73 studies were placebo-controlled, 12 had a β -blocker compared with both placebo and an active substance, and 200 tested a β -blocker against an active comparator. Several of the active-controlled trials compared 2 or more β -blockers, resulting in a total of 361 patient samples exposed to β -blockers (Table S7). Other active comparators or comparator classes used regularly were α 1-blockers, calcium channel blockers (dihydropyridines and nondihydropyridines), blockers of the renin-angiotensin system, and thiazide/thiazide-like diuretics.

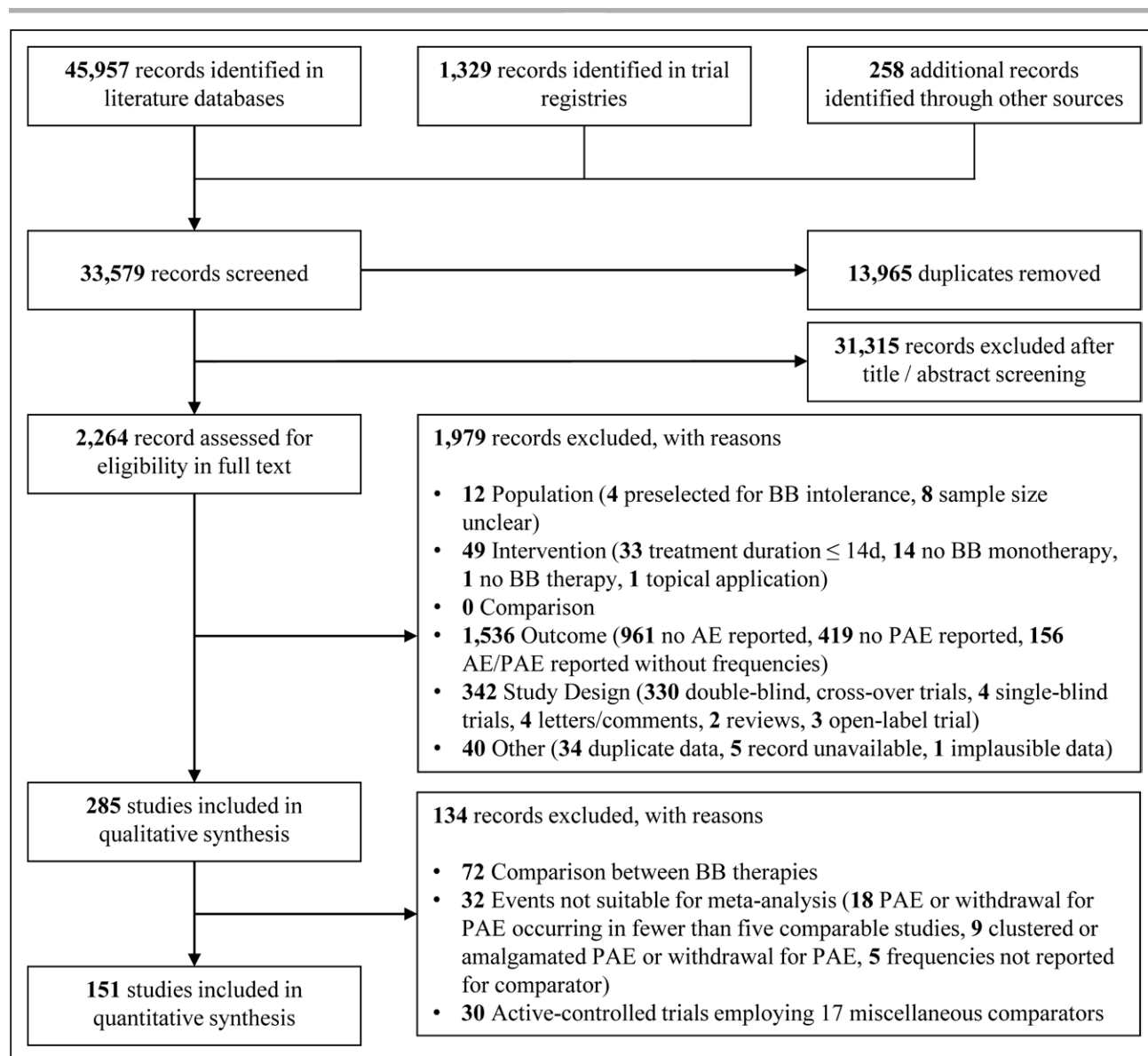


Figure 1. Study selection flowchart.

Results of the database and trial registry search. AE indicates adverse event; BB, β -blocker; and PAE, psychiatric adverse event.

The majority of trials were conducted in patients with cardiovascular diseases, the most frequent indication being arterial hypertension (32 042 patients on β -blockers in 197 trials), followed by myocardial infarction, HF, and angina pectoris. Migraine was the most prominent noncardiovascular indication (1313 patients in 19 trials; Table S8). An overview of the characteristics of each trial included in our review can be found in Tables S9 and S10.

The risk of bias was judged to be high in 225 studies in at least one category (Figure S1). The quality of AE reporting was low in most cases, with only 108 studies reporting all observed AE and only 127 studies reporting AE assessment methodology. Only 53 studies reported using active assessment methods (Figure S2). Altogether, 26 trials (encompassing 5432 patients

on β -blockers) were unpublished. For an additional 21 published trials (12 900 patients on β -blockers), AE data had either not or only incompletely been disclosed before. A total of 157 studies had indicators of industry sponsorship.

Qualitative Analysis

In total, >30 distinct PAEs were reported across the included studies, but no individual symptom was observed in >35% of the studies. Depression was the most frequently reported PAE, with 1600 cases out of 26 832 exposed patients. Similarly, depression was also the most common PAE to cause withdrawal of therapy, with 47 cases out of 13 225 exposed patients.

Quantitative Analysis of Placebo-Controlled Trials

Data gathered from 31 placebo-controlled trials allowed for meta-analytic comparison of β -blocker and placebo treatments for occurrence of depression (Table). The risk for depression was no higher during β -blockers than during placebo treatment (odds ratio, 1.02 [CI, 0.83–1.25]; $P=0.88$; Figure 2). Heterogeneity was low ($P=22\%$). Visual inspection of the funnel plot was not suggestive of publication bias (Figure S3). Similarly, no effect of β -blocker treatment compared with placebo was detected in 9 trials reporting withdrawal due to depression (odds ratio, 0.97 [CI, 0.51–1.84]; $P=0.91$; Figure S4). There was no heterogeneity ($P=0\%$). Results of sensitivity and subgroup analyses were unremarkable (Tables S11 and S12).

Quantitative Analysis of Active-Controlled Trials

Altogether, 27 trials contributed data for meta-analytic comparisons of β -blockers against other active treatments for frequencies of depression (versus nondihydropyridines calcium channel blockers, α 1-blockers, and renin-angiotensin system blockers) or withdrawal for depression (versus dihydropyridines calcium channel blockers). The unadjusted results and sensitivity analyses are compiled in Table S13, and possible subgroup analyses are shown in Table S14. None of the comparisons indicated significant differences between treatments.

Secondary End Points

The results on PAEs other than depression and on the positive control symptom fatigue are presented in detail in the [Data Supplement](#) (Expanded Results 1–3). Briefly, aside from depression, sleep-related PAE (ie, unusual dreams, insomnia, and sleep disturbance) occurred most often, followed by somnolence, loss of appetite, and anxiety. Analyses of placebo- and active-controlled trials indicated possibly increased risks for unusual dreams, insomnia, and sleep disturbance, but a possibly reduced risk for anxiety.

The positive control symptom fatigue/tiredness was reported more commonly than any PAE and was also a frequent reason for the withdrawal of therapy. The risk of fatigue/tiredness was higher during β -blocker therapy than during placebo, dihydropyridine calcium channel blockers, and renin-angiotensin system blockers. Furthermore, patients on β -blockers also had a possibly increased risk of withdrawal for fatigue/tiredness compared with patients on placebo.

DISCUSSION

Our study provides a detailed overview of PAEs occurring during β -blocker therapy in the standardized setting

of double-blind RCTs. Data from 285 studies indicate that PAEs commonly occur during β -blocker therapy, with depression, insomnia, unusual dreams, sleep disturbance, and somnolence being most prevalent. The symptoms detected in our sample mostly mirror those documented in the American and European postmarketing surveillance systems (Table S1). In the past, the emergence of such symptoms has been causally attributed to β -blocker therapy, although this view has been challenged before.^{22,23} Among possible PAEs of β -blockers, depression has been researched most intensively. Luijendijk and Koolman²⁴ reviewed the evidence for and against depression being caused by β -blocker therapy. They noted that several of the studies were underpowered, therefore, failing to provide conclusive evidence for or against an association.²⁴ Studies based on large-scale data are necessary to detect patterns in AEs reliably. Notably, a previous systematic review and meta-analysis done by Ko et al¹⁵ had already indicated a lack of association between β -blockers and depression. Similarly, a recent Danish nationwide population-based register study did not observe adverse associations between β -blockers and incident depression and even found possible protective effects of propranolol, atenolol, bisoprolol, and carvedilol.²⁵ Our meta-analyses of PAE in placebo- and active-controlled studies support and extend these findings by providing substantial evidence against an association of β -blocker use and most PAEs, including depression; with sleep-related disorders (unusual dreams, insomnia, and sleep disorder) being possible exceptions. Furthermore, our analyses suggest that anxiety might occur less frequently in users of β -blockers without intrinsic sympathomimetic activity.

Aside from having the scope on a wider array of psychiatric symptoms, our study differs from the previous work by Ko et al¹⁵ in several key aspects which we consider to be strengths of our study: First of all, due to different inclusion criteria and the inclusion of recent trials, our study sample is considerably larger (28 334 versus 10 662 patients on β -blockers and placebo in the meta-analysis of depression).²⁵ Second, we conducted several sensitivity and subgroup analyses to test our primary findings' robustness and to explore sources of heterogeneity. This included an analysis of the possible effect of industry sponsorship. This is important, as a large fraction of the studies included in our analysis (55%) had clear indicators of industry sponsorship, and industry-sponsored studies more often report favorable outcomes.²⁶ Third, we also tested β -blockers against various active treatments to validate our results. It has been shown that in double-blind trials, physicians can identify patients on β -blockers and placebo to some degree by observing the presence or absence of changes in heart rate.²⁷ This, combined with a possible bias by industry sponsorship, may skew the data collected in placebo-controlled trials in favor of β -blockers. In active-controlled trials, bias

Table. Overview of Studies Contributing Data Towards the Comparison of β -Blockers Versus Placebo for Depression*

Study	Indication, [†] time, and duration	Treatment	Randomized population, N	Analyzed population				Efficacy analysis	Method for adverse event assessment	Quality rating [‡]
				N	♀	Mean age	Race [§]			
Capone, P 1985	HT, ..., and 12 wk	Celiprolol 200–600 mg	...	119	65	51	...	Unclear	...	3
		Placebo	...	80	48	53	...			
Frishman, WH 1994	HT, ..., and 12 wk	Bisoprolol 2.5–40 mg	197	197	...	...	...	Per-protocol	Passive (spontaneous complaints)	3
		Placebo	63	63	...	...	...			
GSK-105517/022	HT, 1988–1989, and 8 wk	Carvedilol 6.25–50 mg	404	404	149	55	NB343, B61	Intention-to-treat	...	3
		Placebo	116	116	43	55	NB98, B18			
Weber, MA 2006	HT, 2004–2005, and 6 wk	Carvedilol 20–80 mg	253	253	86	53	W196, B42, O15	Intention-to-treat	Unclear (questioning)	2
		Placebo	84	84	28	53	W64, B18, O2			
Lewin, A 2013	HT, 2010–2011, and 6 wk	Nebivolol 5–20 mg	290	290	151	51	W177, B106, A3, O4	Intention-to-treat	...	2
		Placebo	143	142	61	50	W92, B46, A3, O1			
Manrique, C 2009	HT, 2001–2003, and 12 wk	Nebivolol 1.25–40 mg	1811	1811	832	54	W1, 320, B466, A15, O10	Intention-to-treat	...	3
		Placebo	205	205	97	53	W134, B71			
Metoprolol trial S-933	HT, ..., and 6 wk	Metoprolol 50 mg	42	42	24	55	...	Intention-to-treat	Active (general and specific questions)	2
		Placebo	42	42	24	58	...			
Moleur, P 1988	HT, 1986, and 4 wk	Bopindolol 0.5–2 mg	86	86	44	53	...	Intention-to-treat	Active (questionnaire)	3
		Placebo	31	31	31	16	...			
Neutel, JM 2010	HT, 2002–2003, and 12 wk	Nebivolol 5–20 mg	502	502	225	53	W265, B149, A8, H74	Intention-to-treat	...	2
		Placebo	167	167	76	54	W93, B48, A1, H25			
Perez-Stable, EJ 1995	HT, ..., and 1 y	Propranolol 40–400 mg	156	156	52	46	W118, O38	Intention-to-treat	Active (checklist)	3
		Placebo	156	156	53	45	W110, O46			
Svetkey, LP 1988	HT, ..., and 10 wk	Nadolol 40–160 mg	51	51	17	53	W43, O8	Unclear	Unclear (questioning)	3
		Placebo	49	49	12	54	W42, O7			
ToMHR 1991	HT, 1985–1994, and > 1 y	Acebutolol 400 mg	132	125	...	...	...	Per-protocol	Active (checklist)	3
		Placebo	234	213	...	...	...			
BEST 2001	HF, 1995–1998, and mean 2 y	Bucindolol 6–200 mg	1354	1354	286	60	W945, B322, H64, O23	Intention-to-treat	...	3
		Placebo	1354	1354	307	60	W951, B305, H79, O19			
Deedwania, PC 2005	HF, 1997–1998, and mean 1 y	Metoprolol 12.5–200 mg	1990	1990	453	64	W1, 870, B106, A7, O7	Intention-to-treat	...	2
		Placebo	2001	2001	449	64	W1, 885, B101, A11, O4			
Goldstein, S 1999	HF, 1994–1996, and 26 wk	Metoprolol 12.5–150 mg	42	42	12	60	W23, B17, A1, O1	Intention-to-treat	...	3
		Placebo	19	19	3	63	W12, B7			
Packer, M 2001	HF, 1997–2000, and 6 mo	Carvedilol 6.25–50 mg	1156	1156	237	63	W1, 049, O107	Intention-to-treat	Unclear (questioning)	2
		Placebo	1133	1133	228	63	W1, 022, O111			
Krum, H 1995	HF, ..., and 14 wk	Carvedilol 25–50 mg	3	33	7	57	...	Intention-to-treat	...	3
		Placebo	16	16	4	53	...			
Sturm, B 2000	HF, 1996–1998, and 24 mo	Atenolol 25–100 mg	51	51	77	51	...	Intention-to-treat	...	3
		Placebo	49	49	5	52	...			

(Continued)

Table. Continued

Study	Indication, [†] time, and duration	Treatment	Randomized population, N	Analyzed population				Efficacy analysis	Method for adverse event assessment	Quality rating [‡]
				N	♀	Mean age	Race [§]			
Wisnibaugh, T 1993	HF, ..., and 3 mo	Nebivolol, 1–5 mg	15	11	6	52	...	Per-protocol	...	3
		Placebo	14	13	6	48	...			
BHAT 1982	MI, 1977–1981, and mean 25 mo	Propranolol 20–240 mg	1916	1916	310	55	W1, 711, O205	Intention-to-treat	Active (specific questions)	2
		Placebo	1921	1921	286	55	W1, 698, O223			
Hansteen, V 1982	MI, 1977–1981, and 12 mo	Propranolol 160 mg	278	278	43	58	...	Intention-to-treat	Passive (spontaneous complaints)	2
		Placebo	282	282	41	59	...			
Julian, DG 1982	MI, 1978–1981, and 12 mo	Sotalol 320 mg	873	873	201	55	...	Intention-to-treat	Unclear (questioning)	3
		Placebo	583	583	99	55	...			
TMTRG 1985	MI, 1982–1984, and 15 d	Metoprolol 200 mg	2877	2877	633	60	...	Intention-to-treat	...	3
		Placebo	2901	2901	661	60	...			
Sotalol Studies 4, 5, 9A, 14	AF, ..., and $\geq$ 6 mo	Sotalol 80–320 mg	415	415	133	62	W379, B16, O20	Intention-to-treat	...	2
		Placebo	282	282	88	63	W252, B9, O21			
Andersson, PG 1983	Migraine, 1982–1983, and 8 wk	Metoprolol 200 mg	...	34	25	42	...	Per-protocol	Active (questionnaire)	3
		Placebo	...	37	35	37	...			
Diener, HC 2003	Migraine, 2001–2002, and 26 wk	Propranolol 20–160 mg	143	142	...	41	...	Intention-to-treat	...	2
		Placebo	143	143	109	40	...			
Pradalier, A 1989	Migraine, ..., and 12 wk	Propranolol 160 mg	40	26	...	...	...	Intention-to-treat	Active (questionnaire)	3
		Placebo	34	22	...	...	...			
Silberstein, SD 2012	Migraine, 2008–2010, and 6 mo	Propranolol 60–240 mg	96	96	84	39	W82, HL5, O9	Intention-to-treat	...	2
		Placebo	95	95	88	42	W81, HL14			
Dransfield, MT 2019	COPD, 2016–2019, and 48 wk	Metoprolol 25–100 mg	268	268	124	65	W178, B83, O7	Intention-to-treat	...	2
		Placebo	264	264	123	65	W194, B60, O10			
Fielding, JF 1981	IBS, ..., and 6 mo	Timolol 30 mg	...	37	20	26	...	Per-protocol	...	2
		Placebo	...	37	24	24	...			
Tchivileva, IE 2020	TMD, 2015–2018, 10 wk	Propranolol 60–120 mg	100	100	77	34	W79, B12, A5, O4	Intention-to-treat	Active (questioning, questionnaire)	2
		Placebo	99	99	78	34	W71, B18, A6, O4			

BEST indicates Beta-Blocker Evaluation of Survival Trial Investigators; BHAT, Beta-Blocker Heart Attack Trial; GSK, GlaxoSmithKline; TMTRG, The Miami Trial Research Group; and ToMHR, Treatment of Mild Hypertension Research Group.

*Missing information is marked as

†Indications include AF indicates atrial fibrillation; migraine; COPD, chronic obstructive pulmonary disease; HF, heart failure; HT, hypertension; IBS, irritable bowel syndrome; MI, myocardial infarctions; and TMD, temporomandibular disorder.

‡Highest risk of bias across all categories of the Cochrane Risk of Bias tool (1, low risk; 2, unclear risk; and 3, high risk).

§Racial distribution of study population (W, White; B, Black; and O, Other [definition varies across studies], A, Asian; H, Hispanic; and HL, Hispanic/Latino).

||Median.

may still be present but is likely more dependent on the study incentive, that is, whether the β -blocker or the other active drug is the focus of the study. Furthermore, we were also able to conduct analyses against nondihydropyridines calcium channel blockers, which may be less distinguishable from β -blockers due to their effect on heart rate. Fourth, we used a well-known AE of β -blockers as a positive control to further validate our findings. Finally, we collected unpublished AE data both from published and unpublished studies to counter a possible publication bias. Published studies have been shown to yield more favorable outcomes than unpublished studies,

and inclusion of unpublished data can help to get a more precise estimate of the true AE risks.²⁸

Several limitations have to be considered when interpreting the results of our study as follows:

First, the majority of β -blocker trials have been conducted >20 years ago. This has several implications: (1) A uniform language to describe AEs had not been established, leading to a plethora of different expressions likely conveying similar PAEs. We tried to amend this by pooling comparable PAEs into the same clusters. (2) In the case of incomplete data, we were unable to contact the original investigators. Second, even in more recent studies,

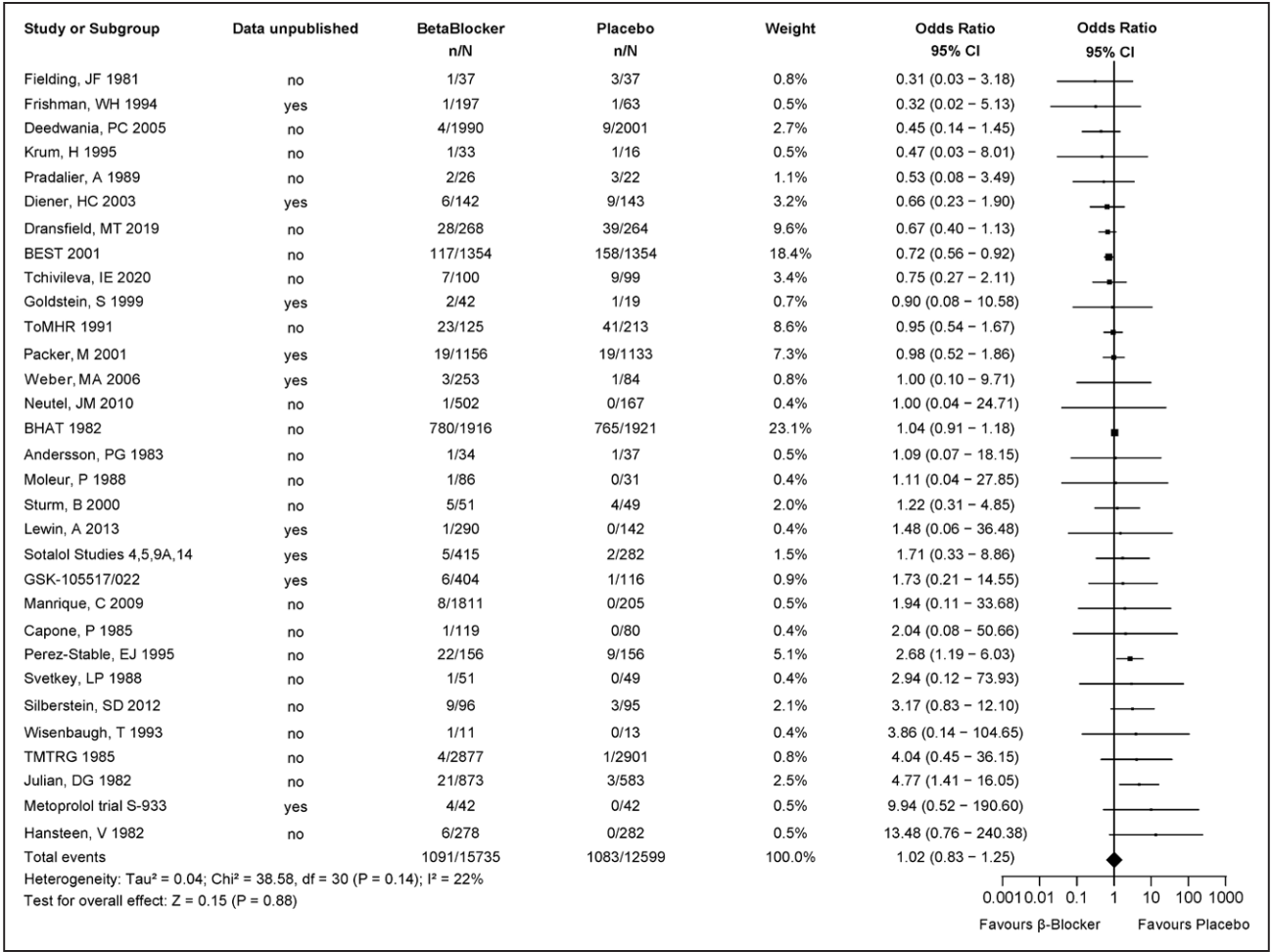


Figure 2. Forest plots for β -blockers vs placebo for depression. BEST indicates Beta-Blocker Evaluation of Survival Trial Investigators; BHAT, Beta-Blocker Heart Attack Trial; GSK, GlaxoSmithKline; TMTRG, The Miami Trial Research Group; and ToMHR, Treatment of Mild Hypertension Research Group.

the absence of an AE report was the prime reason for exclusion despite attempts to gain this information. Third, only 127 studies specified how AEs were assessed, and only 53 studies reported to assess them actively. Assessment methods significantly influence the detection of AE, with passive surveillance methods yielding fewer events than active surveillance.²⁹ Passive surveillance may also be more prone to bias due to the nonstandardized collection of AE data. We performed a sensitivity analysis with restriction to studies using functional assessment to address these issues. Fourth, a considerable number of studies had a high or unclear risk of bias in at least one of the categories of the Cochrane Risk of Bias rating tool. This is particularly important, as the quality of a meta-analysis depends on the quality of its data sources. We had originally intended to control for bias using sensitivity analyses but were unable to implement this, as there was an insufficient number of high-quality studies. Finally, as per our study design, only double-blind, parallel-arm RCTs were eligible. For that reason, open-labeled milestone studies such as the ASCOT (Anglo-Scandinavian

Cardiac Outcomes Trial) could not be included, despite potentially providing valuable data, especially for long-term risks of β -blockers.³⁰

The majority of data sources in our study came from trials investigating β -blockers use in hypertension. The use of β -blockers as first-line therapy in patients with uncomplicated hypertension has been questioned.^{31–33} This concern is reflected in current guidelines for managing of hypertension.² It may thus lead to lower use of β -blockers. However, β -blockers are still important for the therapy of hypertension with cardiac comorbidities, in specific clinical situations, for example, in pregnancy, or in younger hypertensive women planning pregnancy.^{2,4} As depression may increase hypertension-associated cardiovascular morbidity and mortality,¹³ concerns about β -blockers may arise in patients and physicians alike. For that reason, insights about the safety of β -blockers are of great value for the management of hypertension. Furthermore, our results are important for the treatment of HF with reduced ejection fraction, in which β -blockers are indispensable,¹ and also for other cardiovascular

and noncardiovascular indications of β -blocker therapy,³ therefore, affecting current clinical practice across different diseases.

PERSPECTIVES

Our study provides a comprehensive overview of the psychiatric safety of β -blockers across different indications based on large-scale data from double-blind, parallel-arm RCTs. The emergence of PAEs, such as depression, is relatively common in patients requiring β -blocker treatment. Monitoring for such symptoms in clinical practice is undoubtedly justified. However, our data do not support an adverse association between β -blocker use and most PAEs or withdrawal of therapy due to PAEs. Consequently, concerns about a potentially increased risk for PAEs are mostly unwarranted. They should not affect the use of β -blockers in clinical practice. Nevertheless, a potential adverse effect of β -blockers on sleep should not be dismissed and explored in further studies.

ARTICLE INFORMATION

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